



SIMATS ENGINEERING



CSA1508-Cloud Computing and Big Data Analytics

Cloud-Based File Storage and Sharing System

Guided By

Dr. J Chenni Kumaran

PRESENTED BY

A Manvitha devi– 192372386



Introduction



- A Cloud-Based File Storage and Sharing System is an application that stores files securely on the cloud.
- It allows users to upload, download, manage, and share files over the internet.
- Users can access their files anytime and from any device with an internet connection.
- Cloud storage reduces the need for physical storage devices such as USB drives and hard disks.
- It supports easy file sharing through secure links or authorized user access.
- It offers scalability, reliability, and automatic backup of data.



Objective



- To develop a secure cloud-based system for storing and managing files.
- To allow users to upload, download, delete, and organize files easily.
- To enable secure file sharing with authorized users through links or permissions.
- To provide user authentication using registration and login features.
- To ensure data security by implementing access control and encryption.
- To allow users to access files anytime and from any internet-connected device.
- To improve collaboration by enabling users to share files efficiently.
- To provide a reliable and scalable cloud storage solution.



Core Technical Explanation



- **User Authentication**

- Secure user registration and login.
- Only authorized users can access the system.

- **Cloud Storage**

- Files are stored securely in cloud storage.
- Users can access files anytime and anywhere.

- **File Upload**

- Users can upload documents, images, videos, and other files.
- Uploaded files are stored safely in the cloud.



Engineer to Excel

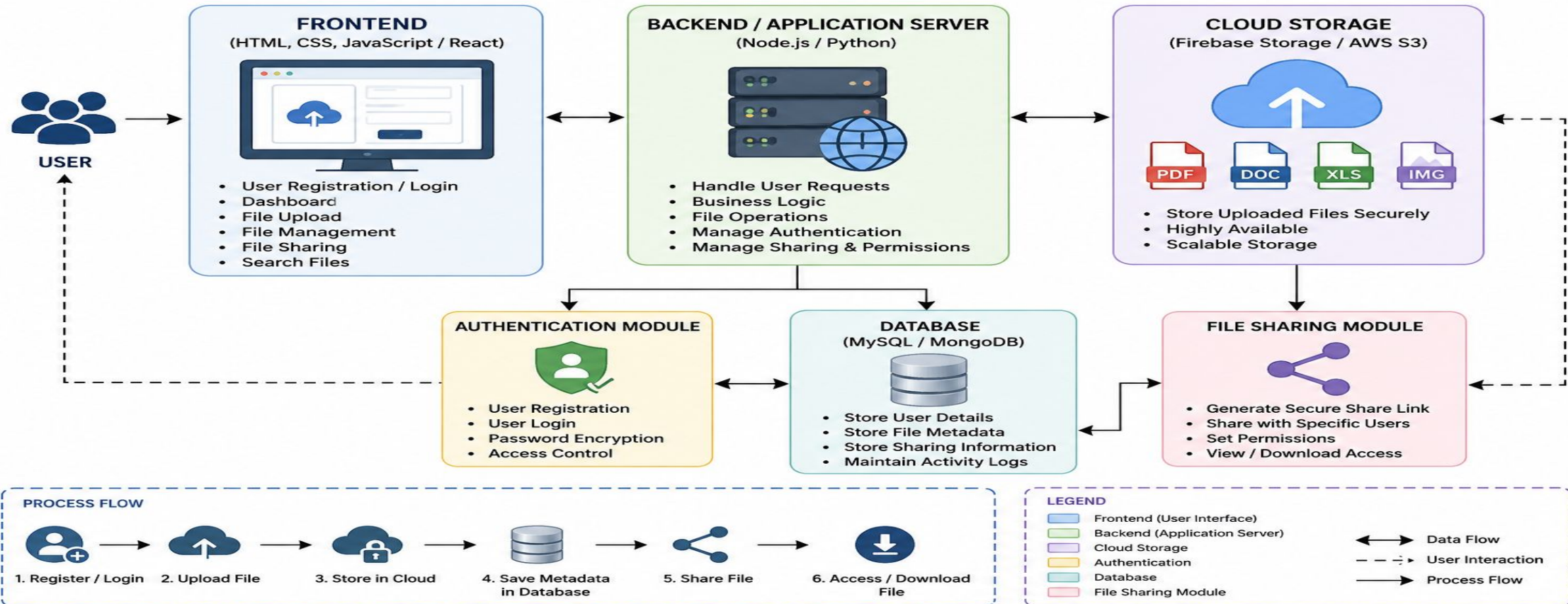
- User Registration
- User Login
- File Upload
- Cloud Storage
- Database Update
- File Management
- File Sharing.
- Access Control
- File Retrieval
- Logout

Process



Architecture

ARCHITECTURE OF CLOUD-BASED FILE STORAGE AND SHARING SYSTEM





Real-World Use Case



- **Educational Institutions:** Students and teachers can upload, download, and share study materials, assignments, and project files.
- **Businesses:** Employees can securely store and share documents, reports, and presentations with team members.
- **Healthcare:** Hospitals can store and share patient records securely among authorized doctors and staff.
- **Government Organizations:** Departments can manage and share official documents safely across different offices.
- **IT Companies:** Developers can collaborate by sharing project files, source code, and software documents.



Real World Application



- **Google Drive** – Stores, manages, and shares files online.
- **Dropbox** – Provides secure cloud storage and file synchronization.
- **Microsoft OneDrive** – Allows users to store and access files across multiple devices.
- **Apple iCloud** – Backs up photos, documents, and device data in the cloud.
- **Educational Institutions** – Share study materials, assignments, and project files with students.
- **Business Organizations** – Store and share reports, presentations, and company documents securely.
- **Healthcare Sector** – Store and share patient records among authorized medical staff.



Advantages



- Easy to access files from anywhere.
- Secure cloud storage for important files.
- Fast file upload and download.
- Easy file sharing with other users.
- Automatic data backup and recovery.
- Reduces the need for physical storage devices.
- Supports multiple file formats.
- Scalable storage based on user needs.



Benefits of cloud



- Access data from anywhere.
- Easy file storage and sharing.
- Low cost and reduced hardware expenses.
- Scalable storage and computing resources.
- High data security. Automatic backup and recovery.
- Fast data access and processing.
- High availability (24/7 access).
- Easy collaboration among users. Supports remote work.



Conclusion



- The **Cloud-Based File Storage and Sharing System** provides a secure and efficient way to store and manage files.
- It allows users to upload, download, and share files from anywhere using the internet.
- The system improves data security through user authentication and access control.
- It reduces the need for physical storage devices and minimizes the risk of data loss.
- Cloud storage provides scalability, reliability, and automatic backup.
- The system supports easy collaboration among users by enabling secure file sharing.

THANK YOU